WoSM Academy

M201-F27 · Calculus II

Power series

Radius of convergence; differentiation and integration term by term

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Last time

- $f_n \to f$ **uniformly** on D if $\sup_{x \in D} |f_n(x) - f(x)| \to 0$.
- Uniform limits of continuous functions are continuous; on $[a, b]$ we may swap lim and $\int$; derivatives need uniform convergence of f_n'.
- **M-test**: $|f_k(x)| \le M_k$ with $\sum M_k < \infty$ gives uniform convergence of $\sum f_k$.
- $\sum x^k = \frac{1}{1-x}$ converges uniformly on every $[-a, a]$ with $a < 1$, so we could integrate and differentiate it term by term.

Today: the same for every power series.

This lecture

01 Power series

02 The radius of convergence

03 Calculus with power series

04 New series from old

05 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Sections 6.1 and 6.2.

01

Power series

Power series

Definition

A **power series centred at** a is a series

$$\sum_{k=0}^{\infty} c_k (x-a)^k = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

with **coefficients** $c_k \in \mathbb{R}$. (We read $(x-a)^0$ as 1, also at $x = a$.)

- A polynomial "of infinite degree". Its partial sums are polynomials.
- It always converges at $x = a$, to c_0.
- For which x does it converge? What function does it define? Can we differentiate and integrate it like a polynomial?

Four examples

With the ratio test (Lecture 3), for $x \neq 0$:

- $\sum x^k$: ratio $|x|$. Converges for $|x| < 1$ only.
- $\sum \dfrac{x^k}{k!}$: ratio $\frac{|x|}{k+1} \to 0$. Converges for every x.
- $\sum k!\, x^k$: ratio $(k+1)|x| \to \infty$. Converges only at $x = 0$.
- $\displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k}$: ratio $\frac{k}{k+1}|x| \to |x|$. Converges for $|x| < 1$ and diverges for $|x| > 1$. At $x = 1$: the harmonic series, divergent. At $x = -1$: the alternating harmonic series, convergent. So: $[-1, 1)$.

In each case the set of convergence is an interval centred at 0 — possibly a single point or all of $\mathbb{R}$.

02

The radius of convergence

Convergence spreads inwards

Lemma (centre $a = 0$; otherwise replace x by $x - a$)

If $\sum c_k x_0^k$ converges for some $x_0 \neq 0$, then $\sum c_k x^k$ converges absolutely for every $|x| < |x_0|$, and uniformly on $[-r, r]$ for every $r < |x_0|$.

Proof.

The terms of a convergent series tend to 0, so they are bounded: $|c_k x_0^k| \leq M$ for all k. For $|x| \leq r$, $|c_k x^k| = |c_k x_0^k| \left|\frac{x}{x_0}\right|^k \leq M q^k$ with $q = \frac{r}{|x_0|} < 1$. As $\sum M q^k$ converges, the M-test applies. □

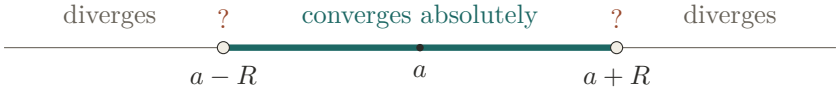
The radius of convergence

Theorem

For every power series $\sum c_k(x-a)^k$ there is an R with $0 \le R \le \infty$, the **radius of convergence**, such that the series

- converges absolutely if $|x-a| < R$, uniformly on every $[a-r, a+r]$ with $r < R$;
- diverges if $|x-a| > R$.

Proof: let R be the supremum of $|x-a|$ over all x where the series converges, and apply the lemma.



Computing R

Theorem

If $c_k \neq 0$ and $\left|\dfrac{c_{k+1}}{c_k}\right| \to L$, then $R = \dfrac{1}{L}$ (with $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$). The same holds if $\sqrt[k]{|c_k|} \to L$.

Proof: the ratio of consecutive terms is $\left|\frac{c_{k+1}(x-a)^{k+1}}{c_k(x-a)^k}\right| \to L\,|x-a|$, which is below 1 if and only if $|x-a| < \frac{1}{L}$. Apply the ratio test (Lecture 3).

Remark

The **Cauchy–Hadamard formula** $R = 1/\limsup_k \sqrt[k]{|c_k|}$ always holds, even when the limits above do not exist. We will not need it.

Examples

- $\displaystyle\sum_{k=1}^{\infty} \frac{(x-2)^k}{k\,3^k}$
  $\left|\frac{c_{k+1}}{c_k}\right| = \frac{k}{3(k+1)} \to \frac{1}{3}$, so $R = 3$: converges for $-1 < x < 5$. At $x = 5$: $\sum \frac{1}{k}$, divergent; at $x = -1$: $\sum \frac{(-1)^k}{k}$, convergent. Interval: $[-1, 5)$.
- $\displaystyle\sum_{k=0}^{\infty} \frac{x^{2k}}{4^k}$
  Every other coefficient is 0, so the formula does not apply. Ratio test on the terms: $\frac{x^2}{4} < 1$ iff $|x| < 2$, so $R = 2$. (Sum: $\frac{1}{1 - x^2/4}$.)
- $\displaystyle\sum_{k=1}^{\infty} k^k x^k$
  $\sqrt[k]{|c_k|} = k \to \infty$, so $R = 0$: it converges only at $x = 0$.

At the endpoints anything can happen

Series	R	**At** $x=1$	**At** $x=-1$
$\sum x^k$	1	diverges	diverges
$\sum \frac{x^k}{k}$	1	diverges	converges conditionally
$\sum \frac{(-1)^k x^k}{k}$	1	converges conditionally	diverges
$\sum \frac{x^k}{k^2}$	1	converges absolutely	converges absolutely

Check the endpoints separately, with the tests of Lectures 2 and 3.

Abel's theorem, without proof

If the series converges at $x = a + R$, its sum is continuous there from the left: $\lim_{x \to (a+R)^-} \sum c_k (x-a)^k = \sum c_k R^k$. Likewise at $a - R$.

03

Calculus with power series

Differentiation and integration term by term

Theorem

Let $\sum c_k (x-a)^k$ have radius $R > 0$ and sum $f(x)$ for $|x-a| < R$. Then f is continuous and differentiable there, and

$$f'(x) = \sum_{k=1}^{\infty} k\, c_k (x-a)^{k-1}, \qquad \int_a^x f(t)\,\mathrm{d}t = \sum_{k=0}^{\infty} \frac{c_k}{k+1}(x-a)^{k+1}.$$

Both new series have the same radius R.

- Inside the interval of convergence a power series behaves like a polynomial.
- The radius stays the same; the behaviour at the endpoints may change.

Proof

Proof.

Take $a = 0$ and $|x| < r < R$. The series converges at r, so $|c_k r^k| \le M$. Then

$$|k c_k x^{k-1}| \le \frac{M}{r}\, k\, q^{k-1}, \qquad q = \frac{|x|}{r} < 1,$$

and $\sum k q^{k-1}$ converges (ratio test). So the derivative series converges absolutely for $|x| < R$, and by the M-test uniformly on every $[-r', r']$ with $r' < R$. Lecture 4: we may differentiate term by term there. For integrals: on $[0, x]$ the series converges uniformly; swap $\int$ and $\sum$. The radius cannot grow: $|c_k x^k| \le |x| \cdot |k c_k x^{k-1}|$ for $k \ge 1$. □

Coefficients are derivatives

Corollary

If $f(x) = \sum c_k (x-a)^k$ for $|x-a| < R$ with $R > 0$, then f has derivatives of every order there, and

$$c_k = \frac{f^{(k)}(a)}{k!} \qquad (k = 0, 1, 2, \dots).$$

Proof: differentiate k times. All terms with $j < k$ disappear; the term with $j = k$ becomes $k!\, c_k$; the terms with $j > k$ still contain a factor $(x-a)$, which is 0 at $x = a$.

- **Uniqueness**: two power series centred at a that agree near a have the same coefficients. However we find a power series for f, it is the same one.
- Next lecture: start from f and use $c_k = \frac{f^{(k)}(a)}{k!}$ — the Taylor series.

Endpoints can change

Example

- $\sum_{k=1}^{\infty} \frac{x^k}{k}$ converges on $[-1, 1)$. Its derivative $\sum_{k=1}^{\infty} x^{k-1}$ converges on $(-1, 1)$ only.
- $\sum_{k=1}^{\infty} \frac{x^k}{k^2}$ converges on $[-1, 1]$. Its derivative $\sum_{k=1}^{\infty} \frac{x^{k-1}}{k}$ converges on $[-1, 1)$.

Differentiating can lose an endpoint; integrating can gain one. The theorem is about the open interval $(a - R, a + R)$.

04

New series from old

Substitution into the geometric series

Start from $\frac{1}{1-u} = \sum_{k=0}^{\infty} u^k$ for $|u| < 1$:

- $u = -x$: $\dfrac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k, \quad |x| < 1.$
- $u = -x^2$: $\dfrac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \quad |x| < 1.$
- $u = \frac{x}{2}$: $\dfrac{1}{2-x} = \dfrac{1}{2} \cdot \dfrac{1}{1-x/2} = \sum_{k=0}^{\infty} \dfrac{x^k}{2^{k+1}}, \quad |x| < 2.$
- Another centre, $u = -(x-1)$: $\dfrac{1}{x} = \dfrac{1}{1+(x-1)} = \sum_{k=0}^{\infty} (-1)^k (x-1)^k,$ $|x-1| < 1.$

Note $\frac{1}{1+x^2}$ is defined for every x, yet its series has $R = 1$. Something at $x = \pm i$ in the complex plane decides that — a story for complex analysis.

Integrating term by term

- $\ln(1+x) = \displaystyle\int_0^x \frac{\mathrm{d}t}{1+t} = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \qquad (|x| < 1)$
- $-\ln(1-x) = \displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k} \qquad (|x| < 1)$
- $\arctan x = \displaystyle\int_0^x \frac{\mathrm{d}t}{1+t^2} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \qquad (|x| < 1)$

Remark

At $x = 1$ the series for $\ln(1+x)$ and $\arctan x$ converge (alternating series). By Abel's theorem their sums are $\ln 2$ and $\arctan 1 = \frac{\pi}{4}$. Next lecture proves both directly, with an estimate of the error.

Differentiating term by term

From $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$, for $|x| < 1$:

$$\frac{1}{(1-x)^2} = \sum_{k=1}^{\infty} k x^{k-1}, \qquad \frac{x}{(1-x)^2} = \sum_{k=1}^{\infty} k x^k, \qquad \frac{x(1+x)}{(1-x)^3} = \sum_{k=1}^{\infty} k^2 x^k.$$

Example (A binary counter)

An increment flips the trailing 1s and one 0: `1011` $\to$ `1100` flips 3 bits. A fraction $\frac{1}{2^k}$ of all values end in `0` and then exactly $k-1$ ones, so the average number of bit flips per increment is $\sum_{k=1}^{\infty} \frac{k}{2^k} = \frac{1/2}{(1-1/2)^2} = 2$.

The exponential function

Let $E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. Its radius is ∞, and term by term

$$E'(x) = \sum_{k=1}^{\infty} \frac{k\,x^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = E(x), \qquad E(0) = 1.$$

Theorem

$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for every $x \in \mathbb{R}$.

Proof: $\big(E(x)e^{-x}\big)' = E(x)e^{-x} - E(x)e^{-x} = 0$, so $E(x)e^{-x} = E(0)e^0 = 1$.

- $x = 1$: $e = \sum_{k=0}^{\infty} \frac{1}{k!}$, so $\sum_{k=1}^{\infty} \frac{1}{k!} = e - 1$ (Lecture 2).
- $x = -1$: $\frac{1}{e} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$, so $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!} = 1 - \frac{1}{e}$ (Lecture 3).

Multiplying power series

Theorem (Cauchy product)

If $\sum a_k x^k$ and $\sum b_k x^k$ both converge for $|x| < R$, then for $|x| < R$

$$\Big(\sum_{k=0}^{\infty} a_k x^k\Big)\Big(\sum_{k=0}^{\infty} b_k x^k\Big) = \sum_{k=0}^{\infty} c_k x^k, \qquad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Both series converge absolutely, so all products $a_j x^j \cdot b_m x^m$ may be added in any order (Lecture 3) — in particular grouped by $j + m = k$. We omit the details.

Multiplying power series: an example

Example

$\dfrac{1}{(1-x)^2} = \left(\sum x^k\right)^2 = \sum_{k=0}^{\infty}(k+1)x^k$, since $c_k = \sum_{j=0}^{k} 1 \cdot 1 = k+1$. The same as by differentiation.

The sequence (c_k) is the *convolution* of (a_k) and (b_k). In Unit 7 this turns counting problems into algebra.

05

Summary and practice

Summary

- A power series $\sum c_k(x-a)^k$ has a **radius of convergence** $R \in [0,\infty]$: absolute convergence for $|x-a| < R$, divergence for $|x-a| > R$. Check the endpoints separately.
- $R = 1/L$ if $\left|\frac{c_{k+1}}{c_k}\right| \to L$ or $\sqrt[k]{|c_k|} \to L$.
- Uniform convergence on every $[a-r, a+r]$ with $r < R$.
- Inside, differentiate and integrate term by term; the radius does not change.
- $c_k = \frac{f^{(k)}(a)}{k!}$: a function has at most one power series around a.
- New series from old: substitute, integrate, differentiate, multiply. Results so far: $\frac{1}{1-x}$, $\ln(1+x)$, $\arctan x$, e^x.

Practice

1. Find the radius and the interval of convergence:

   (a) $\sum_{k=1}^{\infty} \frac{x^k}{k\,2^k}$ (b) $\sum_{k=1}^{\infty} k^2 x^k$ (c) $\sum_{k=1}^{\infty} \frac{(x+1)^k}{k^2}$ (d) $\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$

2. Find a power series for $\frac{x}{1+x^2}$ around 0, and its radius.
3. Find a power series for $\ln(2+x)$ around 0, and its radius.
4. Find $\sum_{k=1}^{\infty} \frac{k}{3^k}$.
5. Use $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ to write π as a series. How fast does it converge?
6. Show that if $\sum c_k x^k$ has radius R, then $\sum c_k x^{2k}$ has radius $\sqrt{R}$.
7. Find $\left(\sum_{k=0}^{\infty} x^k\right)\left(\sum_{k=0}^{\infty} (-1)^k x^k\right)$ as a Cauchy product, and check your answer another way.

Answers

1. (a) $R = 2$, interval $[-2, 2)$. (b) $R = 1$, interval $(-1, 1)$. (c) $R = 1$, interval $[-2, 0]$. (d) $R = \infty$ (ratio of terms $\frac{x^2}{(2k+1)(2k+2)} \to 0$).
2. $x \sum_{k=0}^{\infty} (-1)^k x^{2k} = \sum_{k=0}^{\infty} (-1)^k x^{2k+1}$, $R = 1$.
3. $\ln 2 + \ln\left(1 + \frac{x}{2}\right) = \ln 2 + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k\, 2^k}$, $R = 2$.
4. $\frac{x}{(1-x)^2}$ at $x = \frac{1}{3}$: $\frac{1/3}{4/9} = \frac{3}{4}$.
5. $\pi = 2\sqrt{3} \sum_{k=0}^{\infty} \frac{(-1)^k}{3^k (2k+1)}$. The terms shrink by a factor of about 3 each time: roughly one decimal digit for every two terms.
6. $\sum c_k (x^2)^k$ converges absolutely if $x^2 < R$ and diverges if $x^2 > R$.
7. $c_k = \sum_{j=0}^{k} (-1)^{k-j}$ is 1 for even k and 0 for odd k: the product is $\sum x^{2k} = \frac{1}{1-x^2} = \frac{1}{1-x} \cdot \frac{1}{1+x}$.

Next time

Lecture 06 — Taylor series (Unit 6)

- Which functions are power series? Taylor and Maclaurin series
- Taylor's theorem and estimates of the remainder
- Series for sin, cos, $(1+x)^\alpha$; and, at last, $\ln 2$ and $\frac{\pi}{4}$

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 6.3 and 6.4.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.